Name _____ Surname _____ Date _____

Test: Measurement and laboratory skills

Version B

Duration: 40 minutes. Total: 60 marks. A calculator is allowed; there is no data sheet. Write your answers on this sheet. Give the unit in every answer that needs one. In the multiple-choice questions (10, 12 and 20) only one answer is correct: write its letter in the box. A wrong answer scores the same as no answer. In questions 15 to 20 show your working.

> **Safety net.** The questions marked ✪ are worth 36 marks out of 60, that is 60%: if you answer all of them correctly, you pass.

Part A

1. ✪ Convert 6.4×10^4 mm to m. Give the answer in standard form. **[2 marks]**

Answer: _____ m

2. ✪ With nothing on it, a digital balance reads 0.6 g. A stone placed on the balance makes it read 24.3 g. Write the correct mass of the stone. **[2 marks]**

Answer: _____ g

3. ✪ Write the number of significant figures of each measurement:

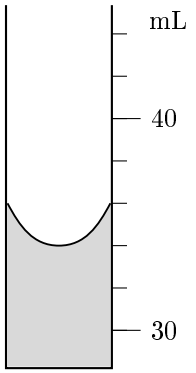
(a) 0.00240 s _____ **[1 mark]** (b) 5.0×10^3 m _____ **[1 mark]**

4. ✪ Convert:

(a) 0.75 kg = _____ g **[1 mark]** (b) 36 mm = _____ m **[1 mark]**

5. ✪ The diagram shows the water in a measuring cylinder. Read the volume of water. **[2 marks]**

Answer: _____ mL



6. ✪ What is the uncertainty of one reading on this cylinder? **[2 marks]**

Answer: ± _____ mL

7. ✪ Write in standard form:

(a) 0.000 074 s = _____ s **[1 mark]** (b) 3 600 000 m = _____ m **[1 mark]**

8. ✪ In the pendulum lab you change the length of the string and measure the time for 20 swings. Write one quantity that must be kept the same (a controlled variable). **[2 marks]**

Answer: _____

9. ✪ Write each number in standard form, with the number of significant figures shown:

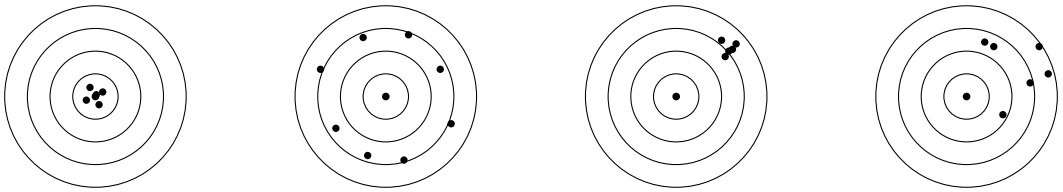
(a) 25 480 m, 2 significant figures: _____ m **[1 mark]**

(b) 0.000 3047 s, 3 significant figures: _____ s **[1 mark]**

10. ✪ Which of these is a derived quantity? **[2 marks]**

A) mass **B)** density **C)** time **D)** length Answer: ☐

11. Four students throw darts. Each dart board stands for repeated measurements of one quantity, and the bullseye is the true value.



1 2 3 4

(a) For each board, write yes or no. **[4 marks]**

Board	Accurate?	Precise?
1		
2		
3		
4		

(b) Write the numbers of the boards that show: **[2 marks]**

a systematic error: _____ a large random error: _____

(c) A student always starts the stopwatch 0.3 s late and stops it exactly on time. Which board shows how his readings look? Is his error random or systematic? **[2 marks]**

Board: _____ Error: _____

(d) Complete the sentence. **[2 marks]**

Taking the mean of many readings reduces _____ error, but not _____ error.

12. In Google Sheets, cells B2 and C2 hold the times for 20 swings of the same pendulum, $t_1 = 18.6$ s and $t_2 = 18.4$ s. Cell D2 is empty. Which formula, typed in E2, gives the mean time? **[2 marks]**

A) `=SUM(B2:D2)` **B)** `=B2+C2+D2/3` **C)** `=AVERAGE(B2:D2)` **D)** `=MAX(B2:D2)`

Answer: ☐

13. Write the SI unit of:

(a) volume _____ **[1 mark]** (b) density _____ **[1 mark]**

14. Round each number to the number of significant figures shown:

(a) 3.14159 s to 3 significant figures: _____ s **[1 mark]**

(b) 0.01996 m to 2 significant figures: _____ m **[1 mark]**

Part B

15. In the density lab a student reads 25 mL before and 42 mL after a stone is put in. Each reading has an uncertainty of half a division (0.5 mL). Calculate the relative uncertainty of the volume of the stone. **[4 marks]**

Answer: _____ %

16. The same stone has a mass $m = 52.7 \pm 0.1$ g and a volume $V = 17 \pm 1$ cm^3.

(a) Calculate the density $\rho = m/V$. Give the unit. **[2 marks]**

Answer: _____

(b) Calculate the relative uncertainty of the density. **[2 marks]**

Answer: _____ %

17. Use your answers to question 16. Calculate the absolute uncertainty of the density and write the density of the stone in the form $\rho \pm \Delta\rho$. Round the uncertainty to 1 significant figure and give ρ to the same number of decimal places. Give the unit. **[4 marks]**

Answer: $\rho =$ _____ $\pm$ _____ _____

18. In the pendulum lab, a group times 20 swings and gets 33.2 s. The uncertainty in starting and stopping the stopwatch is ± 0.2 s. Calculate the time for one swing, with its uncertainty. **[4 marks]**

Answer: $T =$ _____ s $\pm$ _____ s

19. A student times a toy car rolling down a ramp five times: 3.42 s, 3.38 s, 3.40 s, 4.01 s and 3.41 s. The 4.01 s result is anomalous. Calculate the mean of the other four readings. Give your answer to 3 significant figures. **[4 marks]**

Answer: _____ s

20. A pendulum has a period of about 1.6 s, and the uncertainty in the total time is ± 0.2 s. What is the smallest number of swings you must time so that this uncertainty is less than 1% of the total time? **[2 marks for the answer, 2 marks for the reason]**

A) 20 swings **B)** 13 swings **C)** 12 swings **D)** 10 swings Answer: ☐

Why? _____